

Comparison of muscle activation during quadrupedal movement training and traditional bodyweight exercises

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ARTICLE INFO

Keywords:

EMG
Upper extremity
Lower extremity
Trunk muscles
Strength training

ABSTRACT

Quadrupedal movement training (QMT) is a unique form of bodyweight training shown to improve flexibility, FMS scores, and proprioception, however, little is known about its muscle activation characteristics. Therefore, the purpose of this study was to compare surface EMG (sEMG) activity of muscles of the trunk, upper, and lower extremity during QMT movements and common traditional bodyweight exercises (TRA). Twenty (males $n = 9$) participants (age: 27.5 ± 10.8 yrs, height: 169.0 ± 7.0 cm, mass: 66.6 ± 10.9 kg) were prepped and fitted with sEMG electrodes on their right-side bicep brachii, medial deltoid, triceps brachii, pectoralis major, latissimus dorsi, rectus abdominus, rectus femoris and bicep femoris. Following completion of maximal voluntary isometric contraction (MVIC) tests for normalization, participants performed 4 repetitions of 8 QMT exercises (from the Animal Flow™ system) and 3 TRA exercises (pushup, squat and forearm plank) in random order. The pooled peak muscle activations of all QMT exercises resulted in moderate (21–40% MVIC) to very high (>60% MVIC) peak muscle activation levels for each muscle group with significantly higher ($p > 0.05$) muscle activation for the biceps ($47.72 \pm 21.41\%$), deltoid ($66.58 \pm 11.39\%$), and latissimus dorsi ($62.81 \pm 18.14\%$) than the pushup (23.31 ± 16.95 , 45.63 ± 18.59 , $29.17 \pm 20.58\%$ respectively) and significantly higher activation of the bicep femoris ($41.89 \pm 13.83\%$) than the squat ($17.79 \pm 11.12\%$). TRA (pushup) showed significantly higher activation of the triceps and pectoralis. There were no differences in rectus femoris or rectus abdominus. Muscle activation levels during QMT are comparable to or higher than common traditional bodyweight exercises and therefore may be an appealing strategy for training multiple muscle groups simultaneously.

1. Introduction

Quadrupedal movement training (QMT) is a novel form of bodyweight exercise characterized by having the feet and hands in contact with the ground while performing dynamic movements (Buxton et al., 2020; Eckart, 2023). QMT can be used as a stand-alone workout for performance or recovery, integrated into existing programs as part of a dynamic warm-up and/or cool down, or paired with primary resistance training lifts in the body of a workout. QMT has been growing in popularity over the past decade, evidenced by the emergence of numerous training systems throughout the world such as Animal Flow™ (AF) which has been the focus of several recent studies (Buxton et al., 2020, 2022; Eckart, 2023). AF has also grown in popularity over the past decade with now over 1000 certified instructors offering classes and workshops in nearly 50 countries worldwide (Buxton et al., 2022).

The overall body of research regarding QMT is limited. Recent work

from our lab has shown that QMT using AF is a viable alternative form of exercise that meets ACSM exercise guidelines to be considered moderate intensity physical activity (Buxton et al., 2022), and can improve functional movement screen scores, active joint ROM, and dynamic balance following 8 weeks of twice weekly training (Buxton et al., 2020). Additionally, QMT has been shown to improve cognitive function and joint proprioception (Matthews et al., 2016) and trunk muscle activation (Pyka et al., 2017). The growing popularity of QMT and this emerging evidence warrant further studies that would facilitate greater understanding of QMT, including the muscular demands it imposes.

Understanding muscle activation levels during various exercises can help to inform strength and conditioning coaches and practitioners about the intensity of the exercise. Surface electromyography (EMG) has been used as a surrogate measure of intensity due to the previously documented linear relationship between isometric muscle force and EMG amplitude (Milner-Brown and Stein, 1975; Moritani and deVries,

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<https://doi.org/10.1016/j.jbmt.2024.11.002>

Received 1 August 2024; Received in revised form 30 September 2024; Accepted 1 November 2024

Available online 2 November 2024

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1978) and the curvilinear relationship between dynamic muscle contractions and EMG amplitude (Alkner et al., 2000). As such, normalized EMG (nEMG), expressed as a percentage of the maximal EMG amplitude achieved during maximal voluntary isometric contractions (MVIC), can provide a reliable estimate of exercise intensity and is therefore often used for exercise evaluations (Zebis et al., 2019). Furthermore, it is common practice to categorize nEMG data for contextualization accordingly: “low” (<20%), “moderate” (21–40%), “high” (41–60%), and “very high” (>60%) (Kowalski et al., 2022).

Currently, the exercise intensity and muscle activation levels associated with QMT is unknown. As previously mentioned, Pyka et al. observed trunk muscle activation levels using surface EMG recordings during 3 AF movements and found low to very high levels of activation of several trunk muscles (Pyka et al., 2017). These findings provided some initial insight into the intensity of those exercises, however, the exercises observed represent only a fraction of the different types of movements commonly used in QMT and did not address muscle activations of the upper or lower extremities. Therefore, the purpose of this study is to assess trunk, upper, and lower extremity muscle activation during common QMT movements using the AF system and to compare these with common traditional (TRA) bodyweight exercises such as the Pushup, Squat, and Forearm Plank. As such we hypothesized that QMT would result in significantly different activations than TRA. Results from this work will provide greater insight into QMT and how could best be implemented for rehabilitation or training purposes.

2. Methods

2.1. Experimental approach to the problem

A cross-sectional study design with randomized exercise order was used to assess and compare peak muscle activation levels between QMT and TRA exercises. Following a familiarization session all participants were prepped and fitted with 8 wet surface EMG (sEMG) electrodes on the bicep brachii (BB), medial deltoid (MD), long head of the triceps brachii (TB), upper fibers of the pectoralis major (PM), latissimus dorsi (LD), lower fibers of the rectus abdominus (RA), rectus femoris (RF), and bicep femoris (BF) on the right side of their body. After signal acquisition and quality check, all participants performed 2 trials of a MVIC for each muscle group using common manual muscle tests and a handheld dynamometer. Participants then completed in random order four controlled repetitions of eight QMT exercises (taken from the level 1 AF system), a Pushup, a bodyweight Squat, and a Forearm Plank. Peak muscle activation levels (expressed as a percentage of the MVIC) were recorded and analyzed for each muscle during each of the movements. All experimental procedures received local IRB approval prior to implementation (IRB # 111–2023).

2.2. Participants

Twenty (males $n = 9$) healthy volunteers (age: 27.5 ± 10.8 yrs, height: 169.0 ± 7.0 cm, mass: 66.6 ± 10.9 kg) participated in this study (see Table 1). Inclusion criteria included 1) 18–45 years old, 2) currently participating in regular exercise of moderate to vigorous intensity for at

least 30 min, three days per week for at least three months, 3) participation in a form of movement-based exercise (e.g., dance, Yoga, Pilates, QMT, martial arts, etc.), and 4) free of any musculoskeletal injuries that would prohibit or limit physical activity in any way. Exclusion criteria included 1) current smoker, 2) BMI greater than or equal to 30, and 3) known cardio-respiratory, metabolic, or renal disease. Eligible participants were instructed to maintain their normal training and dietary habits throughout the trial, to avoid strenuous physical activity 24 h prior, and food and drink 3 h prior to experimental sessions. During the familiarization session, all participants were oriented to all experimental protocols and made aware of the risks and benefits of participation prior to providing their written informed consent.

2.3. Experimental sessions

Familiarization session. After successful eligibility screening participants visited the laboratory for a familiarization session. All participants were informed of the experimental protocols in addition to the risks and benefits of participation prior to providing their written informed consent. Following consent participants completed a health history form and anthropometric measurements including height (cm), mass (kg), fat free mass (kg) and fat mass (% and kg). Height was measured using a physician’s scale (Detecto, Webb City, MO). Body mass and body composition (fat and lean mass) were measured using a Tanita bioelectrical impedance analyzer (BIA) (MC-980Uplus, Tanita Corporation of America, Arlington Heights, Illinois) following manufacturer procedures. Afterwards all participants were oriented to the surface EMG equipment and procedures, MVIC tests, and each QMT and TRA exercise. Each exercise was practiced at controlled speeds until technical proficiency was observed.

Experimental session. Following the familiarization session participants returned to the laboratory for the experimental session. Participants were prepped and fitted with sEMG electrodes and then performed two trials of each MVIC test for each muscle group. Following MVIC participants completed four repetitions at a controlled speed of the 8 QMT exercises, a Pushup, bodyweight Squat, and a Forearm Plank (10 s hold) in random order. Randomization was conducted using computerized software (www.randomizer.org).

EMG. Prior to electrode placement all participants’ skin sites were prepped by first removing access body hair and then lightly scrubbing the location with a skin prep gel (NuPrep, Weaver and Company, Aurora, CO, USA). Eight dual Ag/AgCl surface electrodes (Noraxon USA, Scottsdale, AZ, USA), with an interelectrode distance of 20 mm were placed on the BB, MD, TB, PM, LD, RA, RF and BF according to SENIAM guidelines and proprietary software (MyoResearch 3.18.10, Noraxon

Table 1
Participant characteristics.

Variable	Total (n = 20, males n = 9)
Age (yrs.)	27.5 ± 10.8
Height (cm)	169.0 ± 7.0
Mass (kg)	66.6 ± 10.9
BMI (kg/m ²)	24.7 ± 5.3
Body fat (%)	18.6 ± 6.7
Fat free mass (kg)	54.4 ± 9.7
Fat mass (kg)	12.6 ± 5.3
Experience (yrs.)	2.5 ± 3.6



Fig. 1. Placement of sEMG electrodes and wireless transmitters for the biceps brachii, medial deltoid, long head of triceps brachii, pectoralis major, latissimus dorsi, rectus abdominus, rectus femoris and bicep femoris.

USA, Scottsdale, AZ, USA) recommendations (Hermens et al., 2000) (Fig. 1.). A wireless EMG sensor system (Ultium, EMG, Noraxon USA Inc., Scottsdale, AZ, United States) with a sampling frequency of 2000 Hz was used to collect the raw sEMG signals. Raw signals were captured and processed in the MyoResearch software where they were filtered with low and high cut-off frequencies of 10 and 500 Hz respectively, rectified and smoothed using root mean square filter (100ms time constant) and then normalized to the peak amplitudes achieved during MVIC testing to be expressed as a percentage of the MVIC. A high-definition camera (Ninox 300C, Noraxon USA, Scottsdale, AZ, USA) sampling at 30 Hz was synched to the sEMG recordings via the MyoResearch software for analysis purposes.

MVIC tests. After placement of all electrodes and signal quality check all participants completed MVIC tests for the BB, MD, TB, PM, LD, RA, RF and BF using common manual muscle testing positions (Kendall et al., 2005) and a handheld dynamometer (Lafayette 01165 A, Lafayette Instrument, Lafayette, IN, USA). Specifically, for BB participants were supine with the elbow flexed to 90° and the dynamometer placed at the distal forearm; for the MD participants were seated and the humerus was abducted to 90° while the elbow was flexed to 90° and the dynamometer was placed at the distal humerus just above the elbow; for the TB the participant was supine with the shoulder flexed to 90° and the elbow flexed to ~45° and the dynamometer placed on the distal forearm; for PM the participant was supine with the shoulder flexed to 90° and the dynamometer placed on the medial aspect of the distal humerus; for the LD the participant was prone with the shoulder extended ~10–20° and the dynamometer placed on the distal forearm; for the RA participants were in a supine position with knees bent and trunk at ~40–45° and the dynamometer placed at the upper sternum (legs were braced by examiner); for the RF participants were seated with the knee flexed to ~70° and the dynamometer placed anteriorly just above the ankle, and for the BF participants were prone with the knee flexed to 90° and the dynamometer placed posteriorly just above the ankle. Participants completed one trial for each muscle group, resting 30s between each test, before completing a second trial of each test. The highest amplitude achieved was recorded and used for normalization.

QMT and TRA exercises. Following MVIC testing participants completed four repetitions of the eig3. ght QMT exercises and three TRA exercises in random order. The QMT exercises represent common movements found in the Level 1 AF system and include 1) Side Kick Through, 2) Full Scorpion, 3) Front Step Through, 4) Crab Reach, 5) Forward Traveling Crab, 6) Forward Traveling Beast, 7) Lateral Ape, and 8) Underswitch. The three TRA exercises include 1) Pushup, 2) body-weight Squat, 3) Forearm Plank (held for one 10s repetition). Illustrations and descriptions of each QMT exercise can be found in [supplemental material 1](#).

2.4. Statistical analysis

Statistical analyses were performed using SPSS version 28.0 (SPSS Inc., Chicago, IL) with statistical significance set *a priori* at $p < 0.05$. Data was tested for normality using the Shapiro-Wilk test. Descriptive statistics (means and standard deviations) for all peak muscle activation levels and demographic variables were computed. Average peak muscle activation achieved during the eight QMT exercises were pooled and compared to TRA. Specifically, the pooled QMT muscle activation of the upper body (BB, MD, TB, PM & LD) was compared to the Pushup, while the pooled data for lower body muscles (RF and BF) were compared to the Squat, and the pooled data for the RA was compared to the Forearm Plank. Dependent t-tests were used to compare differences between pooled QMT data and TRA. Sub-analysis of individual differences in peak activation levels between all exercises performed was assessed using a repeat measures ANOVA and Bonferroni post hoc assessments. The assumption of sphericity was confirmed using Mauchly's test and Greenhouse-Geisser epsilon corrections were applied when the assumption of sphericity was violated. Effect sizes were calculated using

Cohen's d (d : small = 0.2, medium = 0.5, large ≥ 0.8) and partial eta squared (η_p^2 : small = 0.01, medium = 0.06, & large = 0.14).

3. Results

Paired samples t-tests between the pooled peak muscle activations of the 8 QMT exercises were compared against the Pushup for each upper body muscle (BB, MD, TB, PM, LD), the Forearm Plank for the RA, and the Squat for each lower body muscle (RF, BF) (Fig. 2).

3.1. Upper extremity muscles

QMT resulted in significantly greater peak activation of the BB (47.72 ± 21.41 vs. $23.31 \pm 16.95\%$, $p < 0.001$, $d = 1.86$), MD (66.58 ± 11.39 vs. $45.63 \pm 18.59\%$, $p < 0.001$, $d = 1.66$), and LD (62.81 ± 18.14 vs. $29.17 \pm 20.58\%$, $p < 0.001$, $d = 1.34$) than TRA (the Pushup). TRA resulted in significantly greater activation of the TB (73.59 ± 39.47 vs. $48.06 \pm 15.76\%$, $p = 0.004$, $d = 0.79$) and the PM (89.83 ± 23.50 vs. $29.04 \pm 14.33\%$, $p < 0.001$, $d = 2.99$) than QMT.

3.2. Trunk muscle

There was no difference in RA activation between QMT and TRA (Forearm Plank) (57.34 ± 21.13 vs. $61.09 \pm 27.47\%$, $p = 0.527$, $d = 0.15$).

3.3. Lower extremity muscles

There was no difference in RF activation between QMT and TRA (Squat) (62.92 ± 15.71 vs. $62.13 \pm 22.29\%$, $p = 0.865$, $d = 0.04$). QMT resulted in significantly greater peak activation of the BF (41.89 ± 13.83 vs. $17.79 \pm 11.12\%$, $p < 0.001$, $d = 1.58$).

3.4. Exploratory sub-analyses overview

Table 2 shows the peak muscle activations for each muscle group during each exercise. Comparisons using RMANOVA between all exercises as well as exploratory gender analysis can be found in supplemental materials 2. QMT exercises resulted in moderate to very high peak activation for all muscle groups. In general, for all muscle groups except for the PM, a QMT exercise elicited the highest peak activation. Additionally, females tended to show higher activation levels.

4. Discussion

This is the first study to quantify and compare muscle activation levels of QMT exercises to traditional bodyweight exercises. Overall, QMT resulted in medium to very high peak activation levels. The peak activation levels of the BB, MD, and LD from the pooled average of the 8 QMT exercises were significantly greater than the pushup. Additionally peak activation levels of the BF were significantly higher with QMT than during the SQ. There were no significant differences between QMT and TRA in peak activation of the RA and the RF. Muscle activation of the TB and PM were significantly higher in the Pushup.

There were no significant differences in RA activation between QMT (~57%) and the Plank (~61%). The FST resulted in the highest peak RA (~87%) activation of all the exercises performed with the Side Kick Through, Full Scorpion, Crab Reach, Forward Beast, Lateral Ape, and Underswitch producing similar activation levels as the Plank. Pyka et al. were one of the first to investigate core muscle demands during crawling tasks commonly seen in QMT. Here the authors assessed the muscle activation levels of the RF, RA, external oblique and erector spinae during 3 versions of 4-point crawling (referred to as Beast, Beast Activation, and Forward Beast within the Animal Flow system). Trunk muscle activation tended to increase as the stability of the crawling pattern decreased, with RA activity ranging between 18 and 26% (Pyka

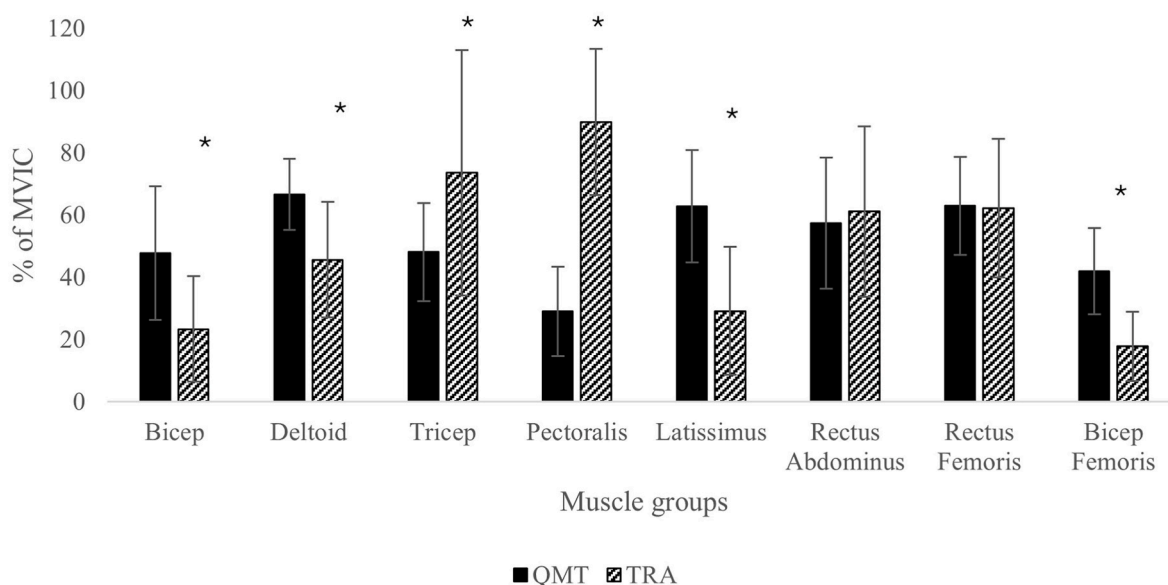


Fig. 2. Pooled peak muscle activation levels of QMT exercises vs. TRA bodyweight exercises (expressed as % of MVIC). QMT = quadrupedal movement training, TRA = traditional, MVIC = maximal voluntary isometric contraction, * = significant difference ($p < 0.005$).

Table 2

Peak muscle activation levels (% MVIC) for each muscle group during QMT and TRA exercises.

	BB	MD	TB	PM	LD	RA	RF	BF
SKT	41 ± 23	64 ± 17	52 ± 23	26 ± 15	71 ± 29	53 ± 23	74 ± 23	44 ± 24
FS	62 ± 28	76 ± 11	46 ± 18	26 ± 15	73 ± 30	53 ± 27	59 ± 21	46 ± 19
FST	46 ± 26	78 ± 17	70 ± 30	29 ± 11	71 ± 35	87 ± 27	87 ± 22	59 ± 29
CR	39 ± 24	75 ± 17	39 ± 19	24 ± 18	63 ± 23	38 ± 28	24 ± 8	43 ± 19
Fcrab	27 ± 12	35 ± 21	40 ± 17	18 ± 13	76 ± 26	31 ± 14	22 ± 9	33 ± 12
Fbeast	23 ± 17	48 ± 16	39 ± 19	25 ± 14	22 ± 12	49 ± 23	76 ± 18	21 ± 15
LA	54 ± 20	84 ± 15	40 ± 17	31 ± 13	36 ± 25	55 ± 32	83 ± 27	29 ± 17
US	46 ± 27	57 ± 16	56 ± 24	28 ± 18	72 ± 31	53 ± 25	68 ± 24	51 ± 24
PUP	18 ± 11	45 ± 18	70 ± 41	89 ± 25	32 ± 23	58 ± 33	24 ± 9	15 ± 13
SQT	4 ± 3	11 ± 6	6 ± 3	7 ± 5	12 ± 5	16 ± 14	60 ± 22	18 ± 11
Plank	16 ± 16	37 ± 20	18 ± 13	14 ± 7	15 ± 10	57 ± 28	28 ± 19	3 ± 2

MVIC = maximal voluntary isometric contraction; QMT = quadrupedal movement training; TRA = traditional; BB = biceps brachii; MD = medial deltoid; TB = triceps brachii; PM = pectoralis major; LD = latissimus dorsi; RA = rectus abdominus; RF = rectus femoris; BF = bicep femoris; SKT = side kick through; FS = full scorpion; FST = front step through; CR = crab reach; Fcrab = forward traveling crab; Fbeast = forward traveling beast; LA = lateral ape; US = underswitch.

et al., 2017). The pooled activation levels of RA during QMT were slightly higher (~57%) which may be due to greater overall stabilization demands of the QMT exercises compared to the crawling patterns used by Pyka et al. During QMT, peak activation of the RA was mostly achieved during times of reduced stability (with either 2 or 3 points of contact). In a similar design to ours, Escamilla et al. compared 10 non-traditional abdominal exercises (e.g., Power Wheel Rollout, Ab Revolutionizer Double Crunch, Hanging Knee-ups, etc.) to 2 traditional ones (Crunch and Bent Knee Sit-up) and found greater trunk muscle activation from the non-traditional exercises. The Crunch and Bent Knee

Sit-up resulted in ~30% and 48% peak activity of the RA (Escamilla et al., 2006). The Power Wheel Rollout resulted in the highest peak lower abdominal activation of ~81% which is comparable to what we found in the Front Step Through (Escamilla et al., 2006). Additionally, Escamilla et al. noted higher activation of secondary muscles like the RF, lumbar paraspinals, and LD during the non-traditional exercises which is consistent with our findings for QMT. Collectively these results suggest that QMT may be similar or superior to traditional abdominal exercises and present an appealing option for individuals with limited training time.

Our findings with respect to muscles of the upper and lower extremities were consistent with those of the trunk, in which QMT resulted in similar or higher activation levels to other common and not-so common exercises. In the present study, muscle activation of the RF and BF during QMT were compared to the bodyweight SQ with no difference between conditions for RF activation (QMT ~63% vs. TRA ~62%) and higher BF activation during QMT (~42% vs. 18%). During most of the QMT movements participants experienced large ranges of motion at the hips and knees while under constant load due to gravity. The Front Step Through resulted in the highest peak activation of RF (~87%) and BF (~59%). During this exercise RF peaked mostly on the return phase of a right leg Front Step Through when the right leg contacted the ground allowing the participant to lower into the Loaded Beast Position, which involves full knee and hip flexion. RF activity was high to very high in all other QMT exercises (aside from Crab Reach and Forward Crab, which primarily load the posterior chain) and resulted in similar activation levels of the RF as those observed by Jakobsen et al. who compared quadriceps activity between total knee arthroplasty rehabilitation exercises requiring equipment (e.g., Leg Press) and versions not requiring equipment (e.g., 1 Leg Squat) (Jakobsen et al., 2019). Regarding the BF, Zebis et al. compared bodyweight exercises used in a validated ACL injury rehab program and found that the single leg Cook Hip Lift produced the highest activity of the BF at ~64% which is higher than we observed overall with QMT but similar to the Front Step Through (Zebis et al., 2019). Interestingly, peak BF activity in the Front Step Through occurred during the return phase when the right leg was performing open chain hip extension reaching the foot back to the start position of the movement. This contrasts with the closed chain nature of the Cook Hip Lift. The Crab Reach is a similar movement pattern to the Hip Lift but uses a bent knee position and both feet in contact with ground which may explain the slightly lower BF activity (~43%) here

than the Hip Lift. Given these similarities in RF and BF activation to more traditional lower body strength and rehabilitation exercises our data suggests the potential for QMT exercises to serve as viable alternative training strategies.

The BB, MD, TB, PM, and LD were evaluated for the upper extremity and QMT exercises were compared with a standard pushup. BB, MD, and LD activation were all significantly greater with QMT than TRA (~48, 67, & 63% vs. 23, 46, & 29%, respectively), while TB and PM activation was greater for TRA (~74 & 90% vs. 48 & 29%). Most likely the higher activation in TB and PM was due to kinematic differences between QMT and the pushup. For each of the QMT exercises performed the elbow remained in a straight or “locked” position resulting in an isometric contraction of the TB but limiting its ability to contract concentrically and eccentrically as it would during the Pushup. Additionally, the humerus experienced horizontal abduction and adduction during QMT, however, the angle between the torso and arms were generally less when compared to the pushup, presumably resulting in less mechanical loading of the PM. Our results, however, showed that the activation levels of the MD, TB and PM during QMT were comparable to unstable pushups and pushups performed on suspension apparatus observed in other studies (Calatayud et al., 2014; Kowalski et al., 2022). Interestingly, QMT resulted in similar PM activation as a barbell bench press in Muyor et al. (~25–30%) while the MD produced levels similar to a Dumbbell Shoulder Press (~62%) and a Seated Lateral Raise (~70%) observed by Sweeney et al. (Muyor et al., 2019; Sweeney, 2014). Of the QMT exercises, Front Step Through resulted in the highest peak activation of TB and PM and the Lateral Ape resulted in the highest MD activation.

Not surprisingly, BB and LD activation were greater for QMT than the Pushup, which again, is most likely due to kinematic differences and therefore limits our comparison. However, in an investigation of muscle activations during the Pullup and several of its variations, Dickie et al. observed high LD activation ranging from ~52 to 58% and very high BB activity ranging from ~81 to 93% (Dickie et al., 2017). LD activation during QMT was slightly higher (~63%) than what was observed in Dickie et al. and 6 of the 8 QMT exercises resulted in very high (>60%) activation. Forward Crab resulted in the highest overall activity with peak occurring most often when the right arm and left leg were supporting the torso as the left hand and right leg lifted and traveled forward. BB activity during QMT was less than (~48%) observed during pullups presumably due to the lack of elbow flexion and extension that occurred during QMT. However, QMT still resulted in high activity of BB with the Full Scorpion producing the highest (~62%). Here, peak activity mostly occurred midway between the fully piked position of the Left Leg Full Scorpion and the end point of the move. During this transition the right arm becomes the primary support along with the right leg while the torso rotates and drops back down to return from its inverted position. This suggests that the BB peaked during its eccentric contribution to this phase of the movement.

As with most studies, it is important to consider potential limitations that may influence the results and interpretation of those results. In the present study both females and males were included and analyzed together. Generally, males tend to demonstrate higher strength to weight ratios than females which may influence activation especially during bodyweight exercises. Exploratory analysis showed that female participants tended to show greater muscle activation than males for the BB, TB, PM, RA, RF and BF, while males exhibited greater activation of the LD in several of the movements performed. Additionally, although all participants met the eligibility criteria for physical activity, several had more experience with QMT than others, and may have also had more experience with resistance training. Clark et al. showed that trunk muscle activation tended to be higher in less experienced participants during barbell squats, suggesting that experience may also influence overall activation levels (Clark et al., 2021). It is unknown how experience may impact activation levels during QMT. Exploratory analysis of our data mostly showed non-significant trends of those with less

experience eliciting higher muscle activation levels than those with more experience. Furthermore, our sample included relatively healthy young, fit individuals reducing the overall generalizability of our findings to other populations. Additionally, although pushups, bodyweight squats, and planks are common bodyweight exercises, many more options exist in which to compare with QMT for a more robust understanding. Finally, although multiple MVIC tests were performed only one test was performed for each muscle group. Research suggests that this may be inferior for establishing MVIC, however, performing more than one test per muscle was not feasible within the given timeframe of the study. Each of these limitations warrants further investigation.

5. Conclusions

Overall, QMT using a system like AF appears to provide a substantial stimulus to upper and lower extremity muscles as well as those of the trunk. Our data suggests QMT is at least comparable and in some cases superior to traditional exercises with respect to muscle activation levels and supports the possibility that QMT may be useful in late-stage rehabilitation programs as well as for general strengthening of the upper and lower extremities and the trunk. Additionally, QMT may be an appealing option for those who have limited training time. More research is warranted to determine appropriate acute training variable (repetitions, sets, tempo, etc.) prescriptions regarding QMT.

Clinical relevance

- QMT is an emerging fitness trend shown to improve movement quality, mobility, and proprioception
- QMT produced moderate to high levels of activation in upper body, lower body and trunk muscles
- Muscle activation during QMT was comparable or superior to traditional bodyweight exercises
- QMT stimulates several muscle groups simultaneously and may appeal to those with limited training time

CRediT authorship contribution statement

Jeff Buxton: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Margaret Daugherty:** Investigation, Data curation, Conceptualization. **Rachel Grubbs:** Investigation, Data curation, Conceptualization. **Makenna Isles:** Investigation, Data curation. **Shanna Mulligan:** Investigation, Data curation, Conceptualization. **Emma Plank:** Investigation, Data curation, Conceptualization. **Elena Taylor:** Investigation, Data curation. **Philip Prins:** Writing – review & editing, Funding acquisition, Conceptualization.

Funding

This project was supported by the Grove City College Swezey fund.

Declaration of competing interest

The authors declare no conflicts of interest.

Acknowledgments

The authors thank all the participants who volunteered their time to take part in this study as well as the Grove City College Exercise Science Department and the college for their support of this project. Funding for this project was secured through the Grove City College Swezey Fund.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbmt.2024.11.002>.

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